

Winter Storm Fern Situation Report

January 2026

Formatted based on LA Fire [Situation Report](#)

Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23rd 2026 and continued through January 26th 2026. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the East Coast. Several states experienced network outages and extremely cold temperatures. For example, temperatures in Seagull, Minnesota and in Iron County, Wisconsin hit lows of -43 degrees and -41 degrees Fahrenheit, respectively ([The Weather Channel \(2026\)](#)). Emergency declarations were issued ¹ ([Congressional Research Service \(2026\)](#)). Power outages due to downed trees and ice occurred in Southern States, such as Texas, Louisiana, Mississippi, and Tennessee. States from Maine to New Mexico experienced significant snowfall, and sleet occurred in the Mid-Atlantic and Northeast. As of January 29th, there were up to 115 fatalities across 20 states after this winter storm, and approximately 2.5 million customers experienced power outages across the United states ([Kothari \(2026\)](#)). Verisk, who are catastrophe risk modelling specialists, estimated a total of US \$4 billion in industry losses with 14 states, ranging from Massachusetts to Texas, that could each exceed \$50 million in insured losses ([Evans \(2026\)](#)).

The following visualizations are developed based on Meta AI for Good's data through Facebook with a focus on Tennessee and Kentucky. The data covers population movement and population density change during the collection period of January 30 to February 12. These datasets use location activity from Facebook users or Facebook Business Pages to estimate how people, population density, network coverage, and business activity change during a crisis, so they do not represent the entire population in the area. Counts of people within a certain region are recorded at 8-hour intervals in this dataset. The counts during the crisis are compared to the counts at a baseline, which are the counts in the same region 45 days prior to data collection. These regions are either Bing tiles, what are about 6 x 6 city blocks, or

¹including Arkansas, Georgia, Indiana, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, and West Virginia

administrative regions as determined by [GADM](#). These counts are aggregated to the county level for this analysis.

The main findings suggest that population-density change during the post-storm period was geographically concentrated rather than evenly distributed across the region. The strongest positive changes (i.e. counties with a population influx at the time) were repeatedly observed in a small number of counties, especially Perry County, Alabama, Mississippi and Cleburne County Arkansas, while Sevier County, Tennessee showed one of the strongest and most persistent negative changes (i.e. counties with an average exodus occurred at the time). Time-series results further suggest that the highest-change counties experienced sustained deviations from baseline, while average- and low-change counties generally stayed closer to zero with short-term fluctuations. However, the average- and low-change counties do have a substantial drop in population density change on February 9th.

How the Data Are Constructed

The Facebook data used in this report comes from Meta’s Data for Good crisis datasets. In plain terms, these datasets use location activity from Facebook users or Facebook business Pages to estimate how people, population density, network coverage, and business activity change during a crisis. The data does not represent everyone in the affected area. For population and movement datasets, it only represents Facebook app users who have Location Services enabled. For business activity, it represents qualifying Facebook business Pages with enough activity to be included while preserving privacy.

The data is constructed by comparing what is observed during the crisis period to what was typical before the crisis. For example, population density during the storm is compared to a pre-crisis baseline of 45-days prior to data collection. Movement during the storm is compared to normal movement patterns between the same places before the event. Business activity is compared to normal posting behavior from business pages before the crisis. Because the data is relative to a baseline, the main signal is not the raw count itself, but whether a place is above or below its pre-crisis level.

Temporally, the population and movement datasets use fixed 8-hour windows. These windows begin at 00:00, 08:00, and 16:00 Pacific Time. This means that the time periods do not automatically adjust to local time zones. For Tennessee and Kentucky, the reported time windows are still based on Pacific Time, which should be kept in mind when interpreting daily patterns. Business activity is reported daily, based on the date value in the dataset.

Some Meta crisis datasets are provided directly as Bing tile data, while others are provided as administrative-region data. **For the rest of this report, we use the Bing tile version as the primary aggregation source because Bing tiles are strictly defined grid cells with clear latitude and longitude coordinates.** This makes the spatial assignment process more transparent and reproducible. In contrast, the administrative-region datasets

are aggregated using boundaries provided to Meta by [GADM](#). Because we do not have direct access to the exact GADM polygons used by Meta for this aggregation, it is harder to verify or reproduce the administrative-region boundaries precisely.

When the data is provided as Bing tiles, each observation is tied to a grid cell with a latitude and longitude. When we add county information to those data, we are not changing the original measurement. Instead, we assign each tile or point to the county that contains it. This allows the tile-level data to be summarized at the county level while preserving a clear record of how the geographic aggregation was performed.

For example, a Bing tile located inside Davidson County, Tennessee would be assigned the Davidson County name and GEOID. If many Bing tiles fall within Davidson County, their values can then be averaged, summed, or otherwise summarized to produce a county-level measure. This approach gives us a consistent way to move from tile-level observations to county-level reporting while relying on publicly available county boundary data.

Facebook Data Definitions

- **Baseline Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the baseline period over users in all tiles in the boundary box defining the crisis
- **Crisis Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the 8-hour window over users in all tiles in the boundary box defining the crisis
- **Population Change Density (%):** Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline (plus a small value, usually 1)
- **Time Windows:** aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00, that represent changes in population density
- **Bing Tiles:** Bing tiles divide the map into a grid of smaller spatial units. Each tile is identified by tile coordinates and a quadkey, which uniquely identifies a tile at a specific level of detail (Microsoft 2024). In the Meta crisis datasets used here, Bing tile level 14 is roughly 2.4 kilometers on each side near the equator, or approximately the size of 6 by 6 city blocks.
- **Administrative Regions:** the polygon boundaries for administrative regions derived from the [Database of Global Administrative Areas](#), also known as GADM (GADM 2022).
- **Percent change:** Percent change in population counts of Facebook app users who have turned on the Location Services device setting on their mobile device and are along the

vector defined by the starting bing tile and ending bing tile, calculated by dividing the difference by the baseline (`n_baseline`) (plus a small value, usually 1).

Aggregation Choices

Imputation and County-Level Percent Change

For the county-level analysis, this report does not use the original tile-level `percent_change` field as the final county-level measure. Instead, the original `percent_change` column is preserved as `given_percent_change`, and a new county-level `percent_change` is calculated from the underlying baseline and crisis counts.

Some rows have missing values for `n_baseline` or `n_crisis` because of Meta’s privacy protections around small counts. To avoid dropping these rows entirely, missing `n_baseline` values are imputed as 3 and missing `n_crisis` values are imputed as 9.

After imputation, observations are summarized by county and reporting window. Because the data may include multiple observations within a single day, the reporting window is defined using both the date and the time-window information available in the data. For each county-window pair, the baseline counts are summed across all Bing tile observations in that county and reporting window, and the crisis counts are summed across those same observations. The county-window percent change is then calculated as:

$$\text{PercentChange}_{cw} = \frac{n_{\text{crisis},cw} - n_{\text{baseline},cw}}{n_{\text{baseline},cw} + 1} \times 100$$

where c is the county and w is the reporting window. The small value added to the denominator follows the logic in the Meta documentation and prevents division by very small baseline values.

This means the maps and tables represent county-level population density change based on aggregated counts.

The aggregation method depends on the type of variable being summarized. For count variables, such as `n_baseline` and `n_crisis`, summing is appropriate because the goal is to estimate the total observed Facebook users across tiles within a county. For relative variables, such as `given_percent_change`, or ratios, averaging can still be useful for descriptive comparisons, but these are not used as the main county-level percent change measure in this report.

For movement data, origin and destination counties are kept separate. Each row represents movement from one starting location to one ending location, so the analysis preserves both `start_county_geoid` and `end_county_geoid`. This avoids collapsing directional movement into a single county and allows county-to-county movement patterns to be analyzed more accurately.

Bing tile observations are assigned to counties using the latitude and longitude coordinates of each tile centroid. The documentation defines these coordinates as the centers of the Bing tile grid cells. These centroid points are spatially joined to county polygons to assign county names and GEOIDs. Quadkeys identify the original Bing tiles, but they are not used directly for county assignment in this report.

Geographically, the data is reported in two main ways: Bing tiles and administrative regions. Bing tiles divide space into small grid cells. At Bing tile level 14, each tile is roughly 2.4 kilometers on a side near the equator, or about the size of several city blocks. Administrative regions instead use boundaries such as counties, states, or other official geographic areas. These are two different approximations of the same general region and same time window, but they can show different rates of population density change (Figure 1).

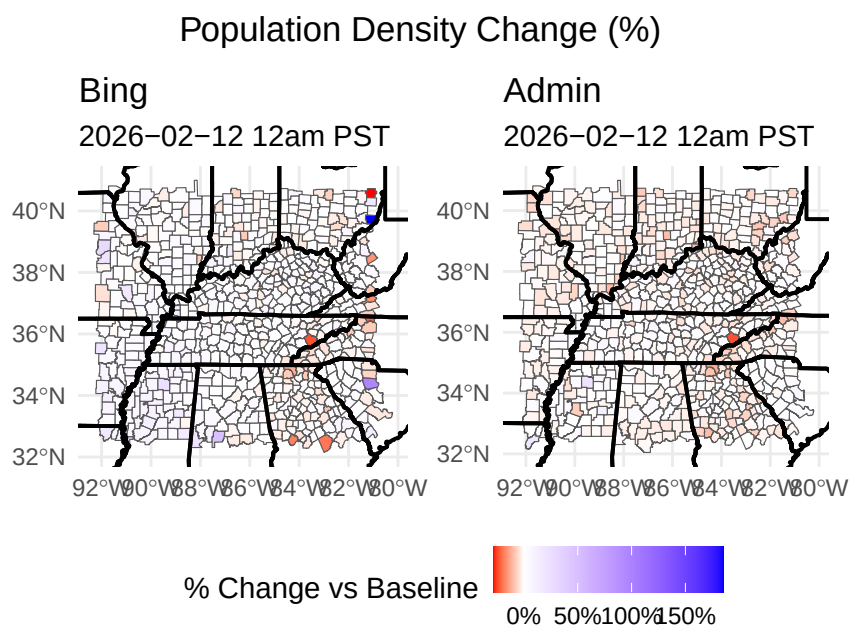


Figure 1: Both maps display county-level summaries for the same region and time window. The bing tile data are aggregated up to counties using tile centroid locations, while the administrative-region data are already provided by Meta at an administrative boundary level.

Takeaways (Figure 1)

- Bing tile results and administrative-region results should not be treated as interchangeable
- The strongest changes appear concentrated in a small number of counties rather than spread evenly across the region

- For example, on February 12th 4pm PST, the population density change for Issaquena, Mississippi is 21.29% based on Bing tiles and only -10.55% based on administrative regions.

Outside Data Added

In addition to the Meta crisis datasets, county-level demographic data was added from the American Community Survey in 2022 (U.S. Census Bureau 2025). This demographic data was incorporated with the use of the `tidycensus` package (Walker and Herman 2025). This outside data provides context about the counties affected by the storm, including total population, median household income, poverty rate, share of residents age 65 and older, and share of households without vehicle access. These variables help interpret which communities may be more vulnerable during a crisis.

The demographic data is joined using `county_geoid`, a standardized county identifier. This is important because county names alone are not unique across states. For example, many states have counties with the same name, but each county has a unique GEOID.

Limitations

These datasets should be interpreted as indicators of crisis-related change, not exact measures of the full population. The population and movement data only include Facebook users with Location Services enabled, so the sample may not represent all residents equally. Areas with fewer Facebook users or lower Location Services usage may be less reliable.

Privacy protections also affect interpretation. Small counts may be suppressed or removed, which can especially affect rural areas. For movement data, this means it is not reliable to calculate exact net flows by simply adding inbound and outbound movement, because some small movement vectors are excluded.

The baseline period is limited to the pre-crisis period available to Meta. This means the analysis does not have a long historical record of pre-crisis behavior. As a result, seasonal changes, holidays, or unusual pre-crisis conditions may affect the baseline. The data is useful for rapid situational awareness, but it should not be interpreted as a complete long-term population or economic dataset.

Finally, the data is time-limited. The documentation notes that some datasets are accessible for 90 days after the final update day and then discarded, which limits longer-term reproducibility and longitudinal analysis.

Movement

% Population Density Change

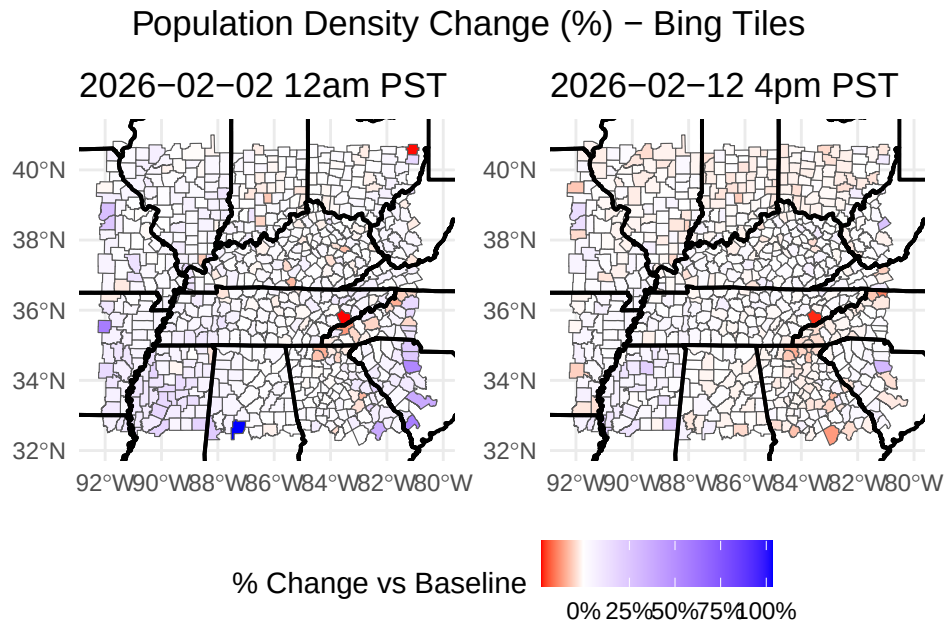


Figure 2: When showing population density change with Bing tiles, there is little negative changes at these two dates.

Takeaways (Figure 2)

- The overall spatial pattern appears fairly stable between the two dates
- There is little visible negative population-density change.
- On February 2nd 12am PST, the counties with the most increased population density are (Table 2):
 - Perry, Alabama (103.0%)
 - Cleburne, Arkansas (58.1%)
 - Hampton, South Carolina (57.4%)
- On February 2nd, the counties with the most decreased population density are (Table 3):
 - Sevier, Tennessee (-22.8%)
 - Carroll, Ohio (-22.7%)

- Swain, North Carolina (-12.4%)
- On February 12th, the counties with the most increased population density are (Table 4):
 - Fairfield, South Carolina (31.3%)
 - Clay, West Virginia (26.2%)
 - Kemper, Mississippi (23.3%)
- On February 12th, the counties with the most decreased population density are (Table 5):
 - Sevier, Tennessee (-21.7%)
 - Alleghany, North Carolina (-14.7%)
 - Laurens, Georgia (-11.8%)

Time Series of Movement

The counties are grouped by average absolute population-density change over the recording period.²

²The percent changes are summarized over the full recording period for each county. For each county, the absolute value of the percent change is taken at each time point, and those absolute changes are averaged across the 14-day period. This creates a single measure of how much each county deviated from baseline, regardless of whether the change was positive or negative. Counties are then compared using this average absolute change to identify the highest-change, lowest-change, and average-change counties. The county-level change measure is defined as $\text{AvgAbsChange}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} |p_{it}|$, where p_{it} is the percent change for county i at time t , and T_i is the number of time periods observed for that county.

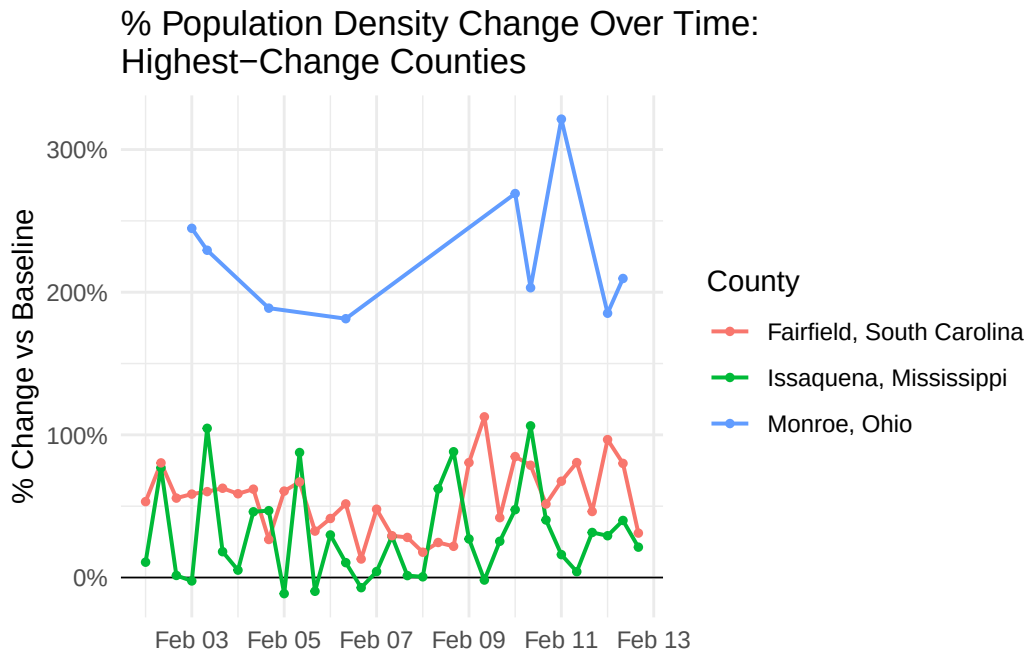
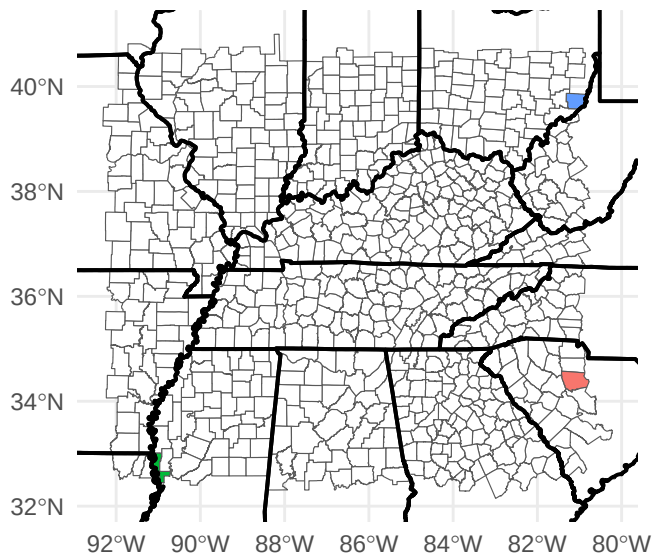


Figure 3: The figure shows that “highest change” includes both large increases and large decreases when counties are selected using largest absolute average percent change in population density. The time period is February 2nd to February 13th at 8hr intervals at 12am PST, 8am PST, and 4pm PST.



Takeaways (Figure 3)

- The time series makes clear that the largest changes are persistent over multiple days,

not just one-day spikes, but each county has starkly different patterns of population density change.

- Monroe has very strong positive population density throughout this entire time period, which is at least double the population density rates of the other two counties.
- Fairfield maintains strong positive population density change throughout this time period.
- Issaquena tends to have peaks around the 8am PST time, which is 10am local time.

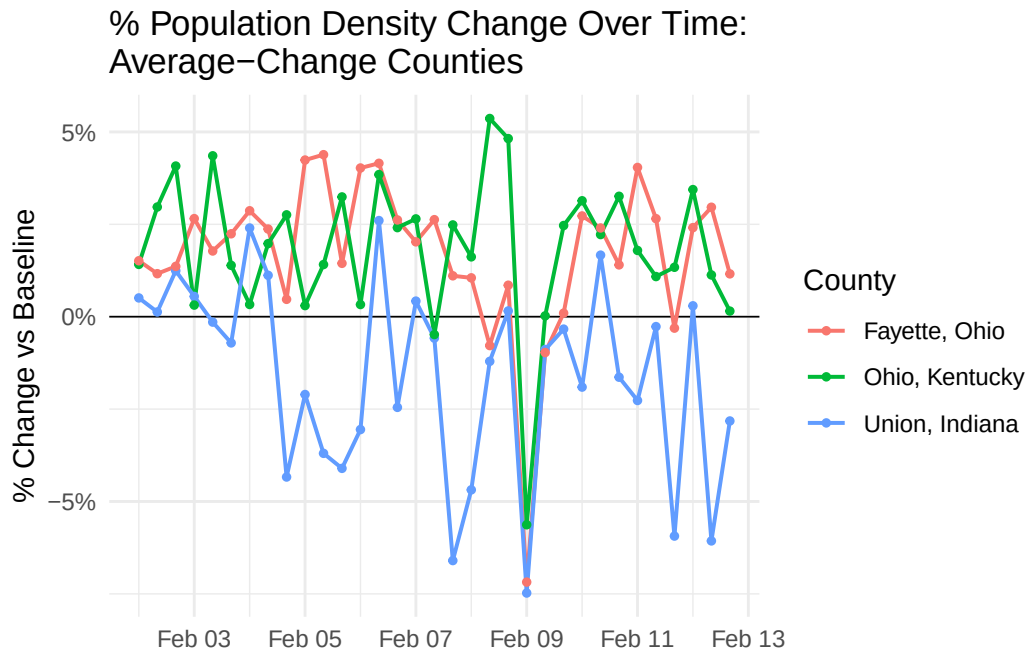
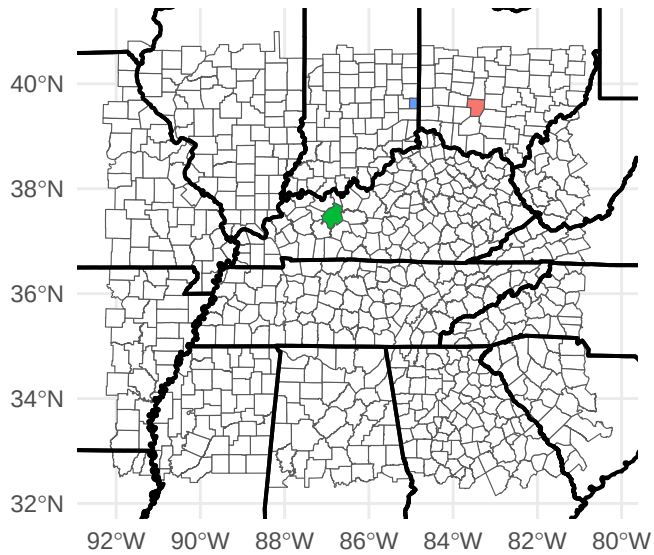


Figure 4: The three counties with average change have more variation in population density change with both positive and negative changes.



Takeaways (Figure 4)

- These counties provide a useful comparison group for interpreting the more extreme counties.
- In comparison to the high-change counties, these three counties have their percentage population density change constrained within $[-10\%, 10\%]$, indicating much lower rates of population density change.
- All counties have a drop in population density change at February 9th of similar magnitudes.
- Fayette and Ohio follow similar trends in variation, but Ohio, Kentucky has a noticeable spike on February 8th.
- Union is the only county of these three that has negative population density rates in this time periods.

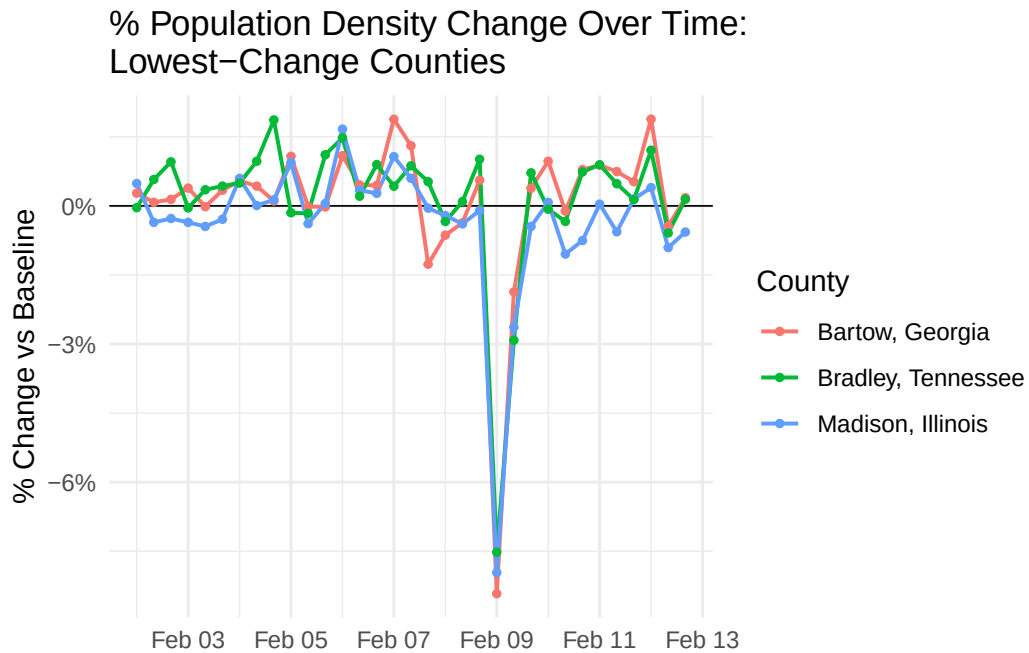
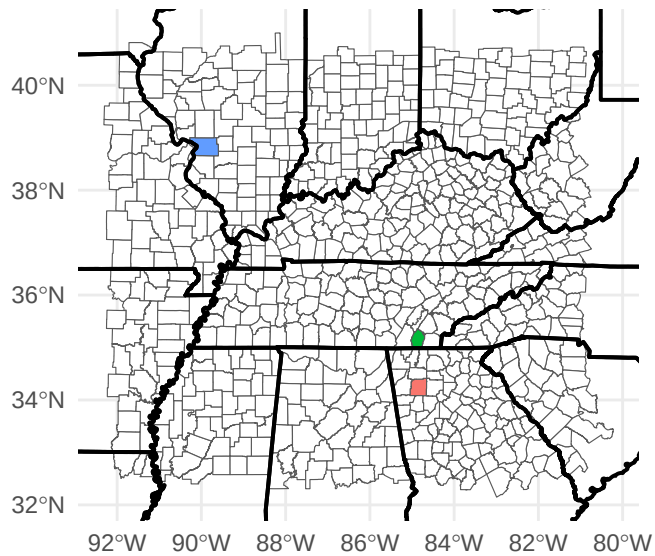


Figure 5: The three counties with the lowest change have a notable drop into extreme negative population density change on February 9th.



Takeaways (Figure 5)

- These three counties follow a similar pattern with relatively stable populations around zero and a sharp drop at February 9th.

- These counties never have strong positive population density change, implying that populations may have either entered before this time period or left after this time period, because the constant negative change indicates people leaving these counties.

Places

Takeaways (Table 1)

- The table is more useful as a data-quality warning than as a substantive movement analysis.
- Because movement vectors with small counts can be suppressed, these low-flow values should not be interpreted as exact zero movement.

Bottom 20 county-to-county crisis movement flows

Source: Bing Tiles

Starting County	Ending County	Total Crisis Movement
Carroll, Georgia	Fulton, Georgia	0
Fulton, Georgia	Sevier, Tennessee	0
Lafayette, Mississippi	Panola, Mississippi	0
Abbeville, South Carolina	Anderson, South Carolina	10
Adair, Kentucky	Russell, Kentucky	10
Aiken, South Carolina	Columbia, Georgia	10
Arkansas, Arkansas	Prairie, Arkansas	10
Attala, Mississippi	Leake, Mississippi	10
Audrain, Missouri	Pike, Missouri	10
Auglaize, Ohio	Shelby, Ohio	10
Banks, Georgia	Habersham, Georgia	10
Barren, Kentucky	Metcalfe, Kentucky	10
Bolivar, Mississippi	Washington, Mississippi	10
Boone, Kentucky	Gallatin, Kentucky	10
Bradley, Tennessee	Polk, Tennessee	10
Buncombe, North Carolina	McDowell, North Carolina	10
Burke, Georgia	Richmond, Georgia	10
Caldwell, North Carolina	Burke, North Carolina	10
Calloway, Kentucky	McCracken, Kentucky	10
Calloway, Kentucky	Vigo, Indiana	10

Table 1: The bottom 20 county-to-county crisis movement flows are all at zero or ten, likely due to data-collection thresholds for privacy concerns. Total crisis movement is the total (non-negative) number of people who have turned on the Location Services device setting on their mobile device for a given starting and ending county.

Counties with Most Increased Population Density on 2026-02-02 12am PST

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Perry	Alabama	103.0	8,031	\$37,654	31.4%	3.3%	8.0%
Cleburne	Arkansas	58.1	25,226	\$55,798	13.9%	3.2%	5.3%
Hampton	South Carolina	57.4	18,254	\$44,711	20.7%	2.2%	11.0%
Fairfield	South Carolina	53.2	20,550	\$47,885	15.6%	3.2%	5.9%
Emanuel	Georgia	37.7	23,026	\$52,772	25.7%	2.8%	8.2%
Orangeburg	South Carolina	37.6	83,253	\$45,675	22.4%	3.1%	9.0%
Callaway	Missouri	28.1	44,741	\$74,176	11.6%	2.2%	5.7%
Osage	Missouri	25.2	13,402	\$76,681	8.6%	2.6%	5.6%
Chester	South Carolina	23.8	32,182	\$52,458	18.7%	2.5%	6.3%
York	South Carolina	22.7	293,673	\$89,095	8.4%	2.2%	3.7%
Sumter	Alabama	22.4	11,878	\$33,310	26.5%	2.7%	9.5%
Kemper	Mississippi	18.9	8,729	\$46,431	24.8%	3.1%	3.4%
Lincoln	Georgia	18.5	7,854	\$56,907	17.0%	4.8%	7.1%
Quitman	Mississippi	18.2	5,774	\$32,412	29.6%	2.4%	10.9%
Madison	Mississippi	17.9	111,647	\$83,441	11.2%	2.2%	5.3%

Table 2: There are a few counties with high population density change but low total population. All demographic information comes from the American Community Survey in 2022. Population density change is the average of the Percent change in Facebook Location Services-enabled user population counts per Bing tile in each county.

Demographic Profile

Counties with Most Decreased Population Density on 2026-02-02 12am PST

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-22.8	99,652	\$62,581	13.5%	2.7%	4.1%
Carroll	Ohio	-22.7	26,659	\$64,835	12.9%	3.5%	6.8%
Swain	North Carolina	-12.4	14,013	\$52,349	19.1%	3.8%	6.6%
Watauga	North Carolina	-9.6	55,123	\$51,693	25.5%	2.3%	4.4%
Fannin	Georgia	-9.5	25,742	\$57,073	13.7%	2.9%	3.0%
Taliaferro	Georgia	-8.5	1,558	\$52,500	20.4%	2.9%	5.8%
Cumberland	Kentucky	-8.2	5,948	\$40,000	24.8%	2.8%	9.4%
Wolfe	Kentucky	-7.5	6,443	\$30,417	37.7%	1.9%	10.3%
Gilmer	Georgia	-7.1	32,426	\$74,499	16.0%	3.2%	2.9%
Brown	Indiana	-6.5	15,606	\$78,528	7.8%	4.5%	3.5%
Pope	Illinois	-6.1	3,739	\$60,050	13.3%	4.1%	5.5%
Ashe	North Carolina	-5.4	26,950	\$56,866	13.5%	3.5%	4.4%
Orange	Indiana	-5.0	19,767	\$65,873	13.4%	3.0%	6.7%
White	Georgia	-5.0	28,810	\$70,666	12.4%	3.2%	2.6%
Buncombe	North Carolina	-4.8	274,360	\$74,436	11.8%	2.5%	4.7%

Table 3: Most of the counties with the lowest population density change have fairly high vehicular access. All demographic information comes from the American Community Survey in 2022. Population density change is the average of the Percent change in Facebook Location Services-enabled user population counts per Bing tile in each county.

Counties with Most Increased Population Density on 2026-02-12 4pm PST

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Fairfield	South Carolina	31.1	20,550	\$47,885	15.6%	3.2%	5.9%
Clay	West Virginia	26.2	7,835	\$42,318	26.4%	3.5%	10.3%
Kemper	Mississippi	23.3	8,729	\$46,431	24.8%	3.1%	3.4%
Issaquena	Mississippi	21.3	928	\$31,429	16.2%	2.7%	14.0%
Lafayette	Mississippi	20.1	58,327	\$67,185	18.5%	2.0%	6.7%
Callaway	Missouri	14.4	44,741	\$74,176	11.6%	2.2%	5.7%
Osage	Missouri	13.9	13,402	\$76,681	8.6%	2.6%	5.6%
Quitman	Mississippi	11.5	5,774	\$32,412	29.6%	2.4%	10.9%
Madison	Mississippi	11.5	111,647	\$83,441	11.2%	2.2%	5.3%
Oktibbeha	Mississippi	11.4	51,771	\$46,695	25.5%	1.7%	5.4%
Sharkey	Mississippi	10.4	3,872	\$40,000	30.9%	2.1%	8.0%
West Carroll	Louisiana	9.5	9,464	\$49,134	16.2%	3.1%	5.9%
Grenada	Mississippi	8.9	21,217	\$48,804	24.6%	2.3%	7.9%
Benton	Mississippi	8.5	7,589	\$46,016	21.5%	2.7%	6.8%
Panola	Mississippi	8.5	32,808	\$45,945	23.3%	2.1%	6.4%

Table 4: Issaquena has the highest population density but notably small population size. All demographic information comes from the American Community Survey in 2022. Population density change is the average of the Percent change in Facebook Location Services-enabled user population counts per Bing tile in each county.

Counties with Most Decreased Population Density on 2026-02-12 4pm PST

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-21.7	99,652	\$62,581	13.5%	2.7%	4.1%
Alleghany	North Carolina	-14.7	11,174	\$47,172	16.0%	2.6%	6.0%
Laurens	Georgia	-11.8	49,798	\$55,010	22.6%	2.3%	6.3%
Fannin	Georgia	-9.9	25,742	\$57,073	13.7%	2.9%	3.0%
Taliaferro	Georgia	-8.6	1,558	\$52,500	20.4%	2.9%	5.8%
Taylor	Georgia	-7.1	7,786	\$41,788	32.1%	4.2%	15.5%
White	Georgia	-6.7	28,810	\$70,666	12.4%	3.2%	2.6%
Ashe	North Carolina	-6.6	26,950	\$56,866	13.5%	3.5%	4.4%
Monroe	Missouri	-6.5	8,734	\$54,516	14.2%	3.6%	7.0%
Gilmer	Georgia	-6.4	32,426	\$74,499	16.0%	3.2%	2.9%
Brown	Indiana	-5.8	15,606	\$78,528	7.8%	4.5%	3.5%
Bland	Virginia	-5.6	6,199	\$61,216	13.6%	1.7%	5.3%
Towns	Georgia	-5.5	12,913	\$59,135	13.3%	3.0%	3.7%
Jefferson	Arkansas	-5.4	64,802	\$51,096	18.6%	2.8%	7.5%
Hancock	Georgia	-5.2	8,650	\$40,082	30.6%	3.6%	8.4%

Table 5: Similar to Feb 2nd, the counties with strong negative population density change also have high levels of vehicular access. All demographic information comes from the American Community Survey in 2022. Population density change is the average of the Percent change in Facebook Location Services-enabled user population counts per Bing tile in each county.

Demographic Correlations

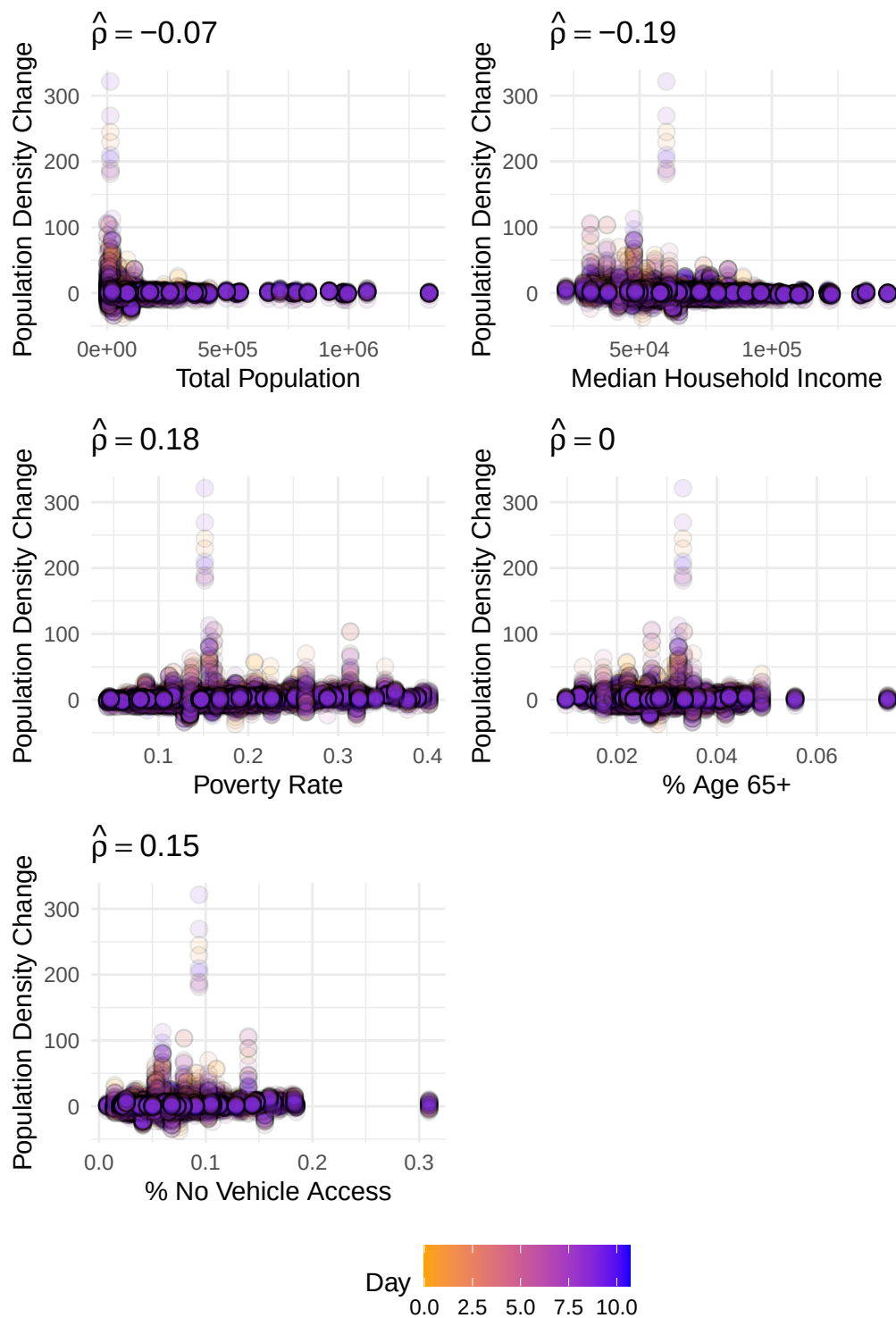


Figure 6: There does not seem to be strong linear correlations between population density change and the other demographic variables. Population density change is the percent change in Facebook Location Services-enabled user population counts per Bing tile in each county at 8-hour intervals. All demographic variables are already at the county level from the American Community Survey in 2022. Days are the 1st to 10th day of data collection, from Feb 2nd to Feb 12. These counties are from 15 Southern states.

Imputation Summary

All available dates and hours

Count	Total Count	Observed Count	Imputed Count	Share of Count Imputed	Total Bing Tiles	Imputed Bing Tiles	Share of Bing Tiles Imputed
Baseline	342,000,436	341,681,836	318,600	0.1%	3,670,271	21,156	0.6%
Crisis	343,836,812	342,572,996	1,263,816	0.4%	3,670,271	21,990	0.6%

Table 6: Share of total baseline and crisis counts and Bing tile observations that come from imputed values for the selected reporting window.

References

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